```
import pandas as pd
```

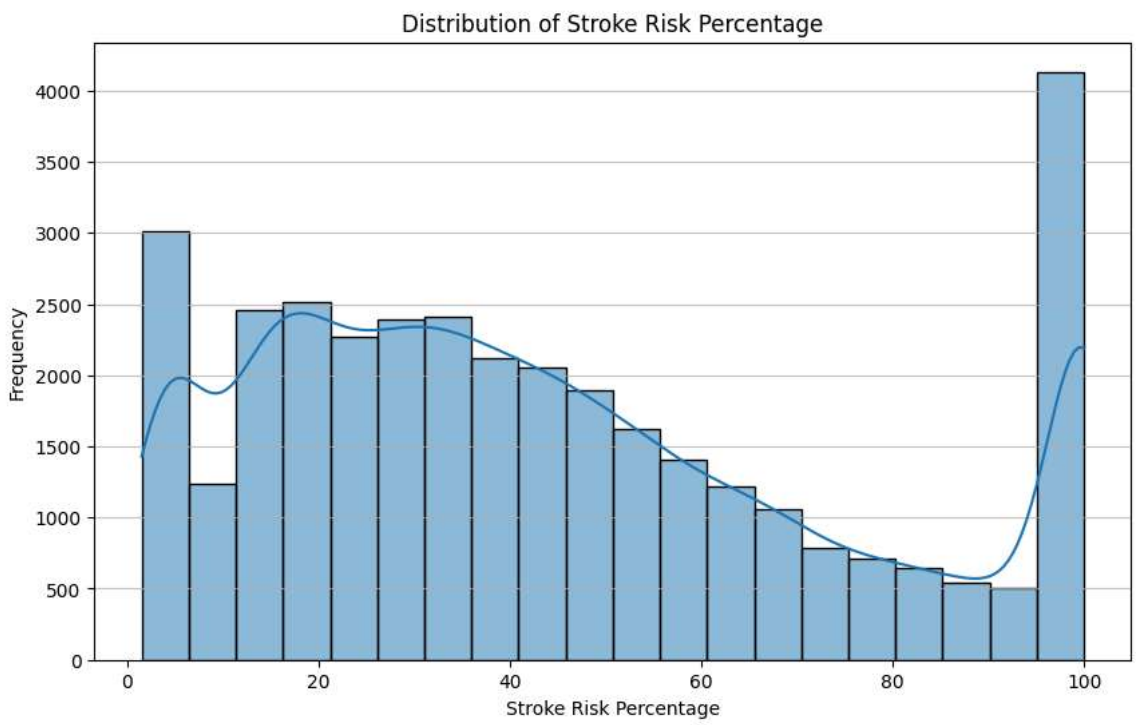
```
path="/content/drive/MyDrive/Dataset/accuracy.data.xlsx"
df=pd.read_excel(path)
df.head(5)
```

	age	gender	chest_pain	high_blood_pressure	irregular_heartbeat	shortness_of_breath	fatigue_weakness	dizziness	swelling_
0	22	Male	1	0	0	0	0	0	
1	52	Male	0	1	1	0	0	0	
2	63	Female	0	1	0	0	0	1	
3	41	Male	0	0	1	0	1	0	
4	53	Male	0	0	0	0	0	1	

Next steps: Generate code with df New interactive sheet

Distribution of Stroke Risk Percentage

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(10, 6))
sns.histplot(df['stroke_risk_percentage'], kde=True, bins=20)
plt.title('Distribution of Stroke Risk Percentage')
plt.xlabel('Stroke Risk Percentage')
plt.ylabel('Frequency')
plt.grid(axis='y', alpha=0.75)
plt.show()
```

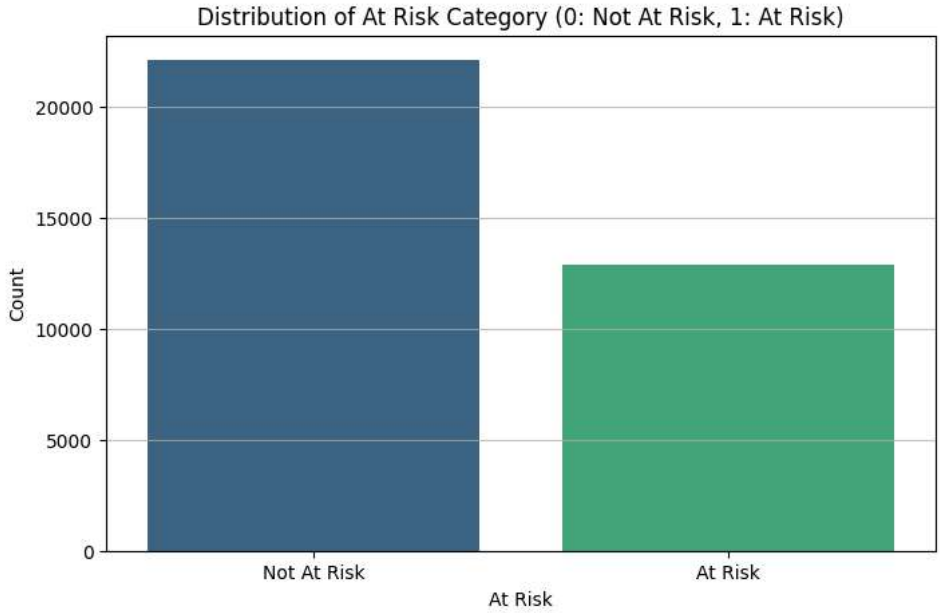


Distribution of 'At Risk' Category

```
plt.figure(figsize=(8, 5))
sns.countplot(x='at_risk', data=df, palette='viridis')
plt.title('Distribution of At Risk Category (0: Not At Risk, 1: At Risk)')
plt.xlabel('At Risk')
plt.ylabel('Count')
```

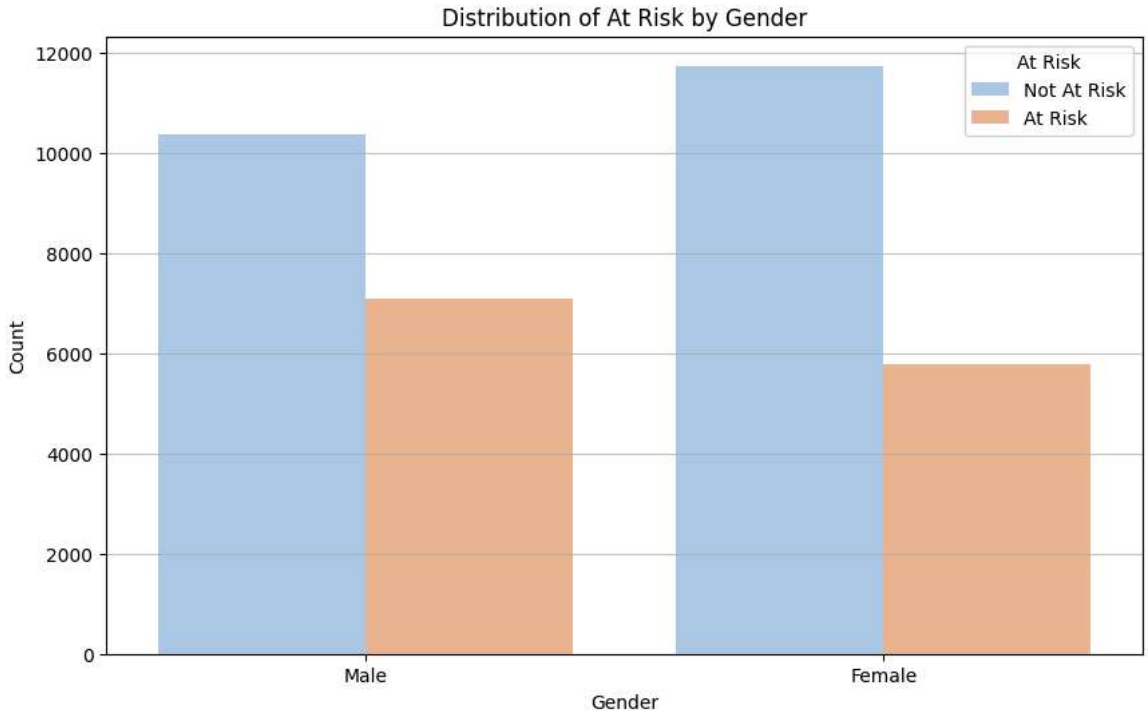
```
plt.xticks([0, 1], ['Not At Risk', 'At Risk'])
plt.grid(axis='y', alpha=0.75)
plt.show()
```

```
/tmp/ipykernel_4958/2411062640.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.countplot(x='at_risk', data=df, palette='viridis')
```



Distribution of 'At Risk' by Gender

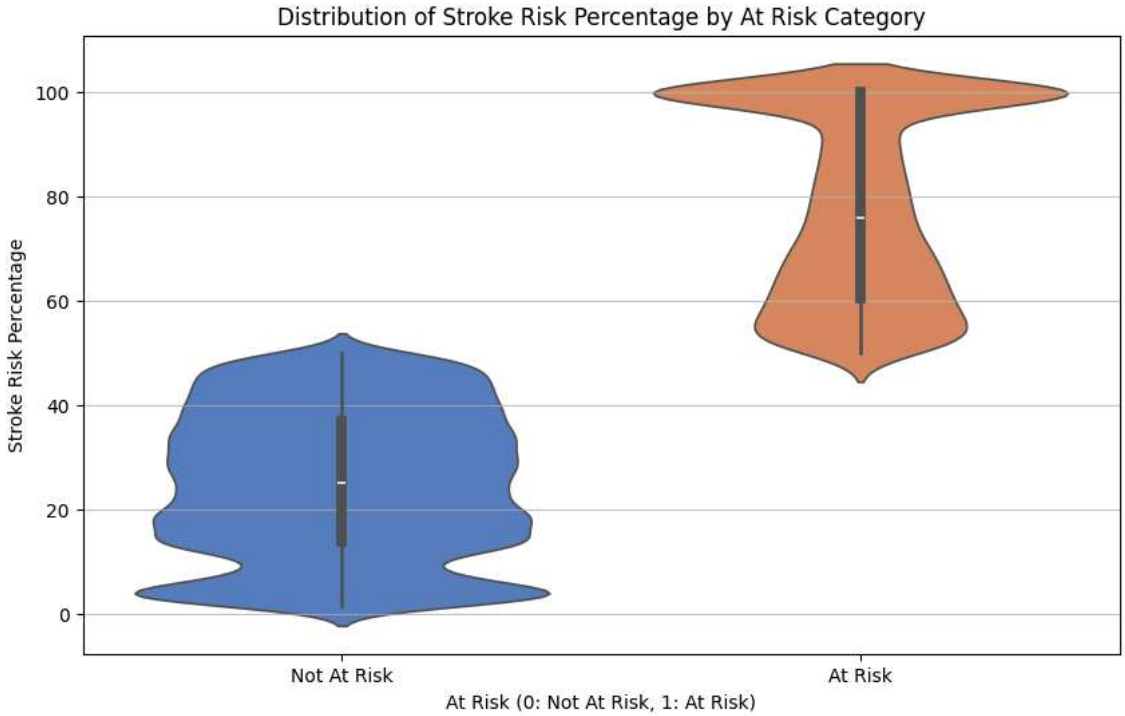
```
plt.figure(figsize=(10, 6))
sns.countplot(x='gender', hue='at_risk', data=df, palette='pastel')
plt.title('Distribution of At Risk by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(title='At Risk', labels=['Not At Risk', 'At Risk'])
plt.grid(axis='y', alpha=0.75)
plt.show()
```



Distribution of Stroke Risk Percentage by 'At Risk' Category

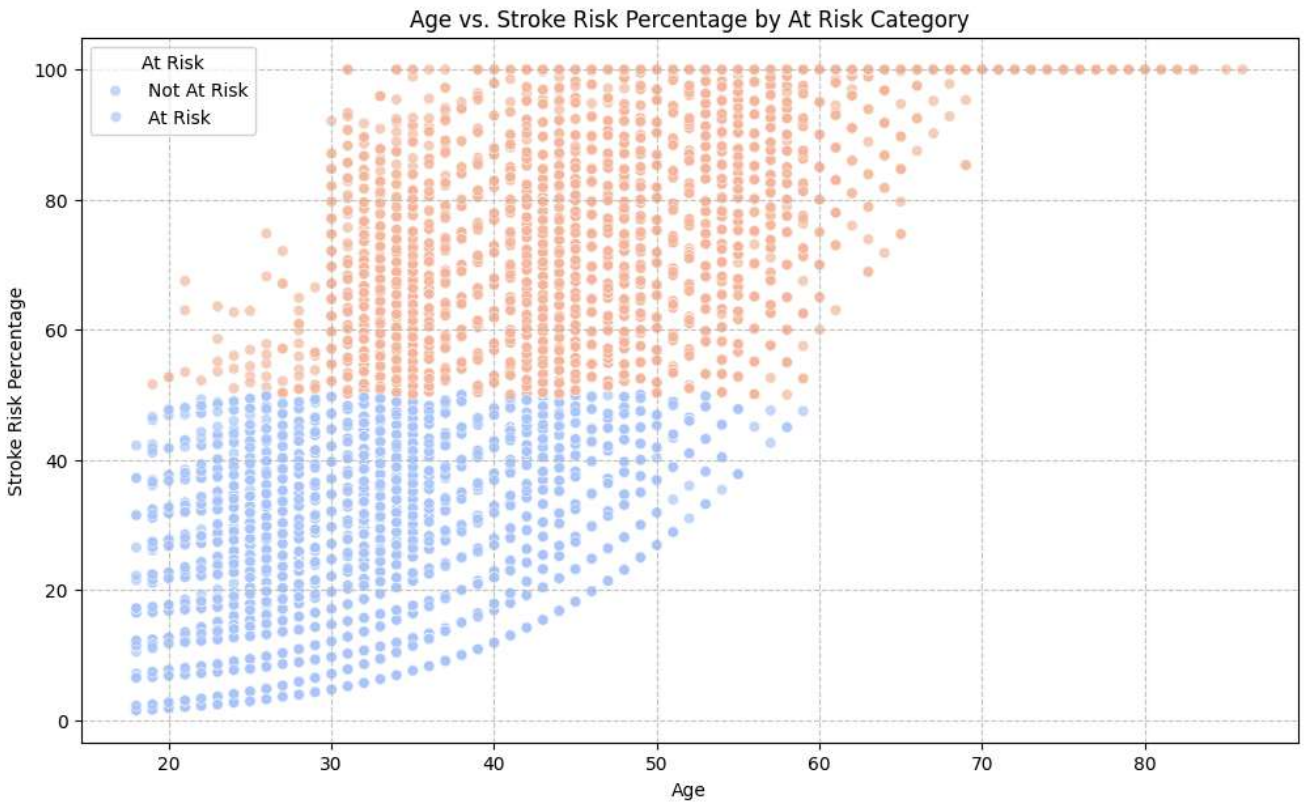
```
plt.figure(figsize=(10, 6))
sns.violinplot(x='at_risk', y='stroke_risk_percentage', data=df, palette='muted')
plt.title('Distribution of Stroke Risk Percentage by At Risk Category')
plt.xlabel('At Risk (0: Not At Risk, 1: At Risk)')
plt.ylabel('Stroke Risk Percentage')
plt.xticks([0, 1], ['Not At Risk', 'At Risk'])
plt.grid(axis='y', alpha=0.75)
plt.show()
```

```
/tmp/ipykernel_4958/2241607911.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.violinplot(x='at_risk', y='stroke_risk_percentage', data=df, palette='muted')
```



Relationship between Age, Stroke Risk Percentage, and At Risk Category

```
plt.figure(figsize=(12, 7))
sns.scatterplot(x='age', y='stroke_risk_percentage', hue='at_risk', data=df, palette='coolwarm', alpha=0.7)
plt.title('Age vs. Stroke Risk Percentage by At Risk Category')
plt.xlabel('Age')
plt.ylabel('Stroke Risk Percentage')
plt.legend(title='At Risk', labels=['Not At Risk', 'At Risk'])
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```



Start coding or generate with AI.